

HyperLocal AI Commerce Platform

System Architecture & Portal Workflows Complete Mapping Guide

This document maps out the complete operational workflow and structural interactions across all 4 primary user portals in the **HyperLocal AI Commerce Ecosystem: Buyer (Customer) Portal, Seller (Merchant) Portal, Delivery Partner (Driver) Portal, and System Administrator (Admin) Console.**

Portal Role	Primary Purpose	Key Technology Features	URL Route
■ Buyer	Local merchant discovery, shopping cart, AI Voice search, & real-time tracking.	AI Voice Copilot, WebSpeech API, Leaflet Maps, Live WebSocket tracking	/#/
■ Seller	Shop profile registration, catalog inventory, orders queue, & flash deals.	2DSphere GeoPin Picker, Real-time Stock Toggle, Flash Sales Broadcaster	/#/seller-dashboar d
■ Driver	Logistics runs, package claims, GPS broadcast, & OTP delivery verification.	Live GPS Coordinates Feed, 4-Digit Security OTP Form, Earnings Wallet	/#/driver-dashboar d
■■ Admin	Platform control console, seller approvals, driver map, & payout ledgers.	Live Logistics Monitor, Merchant Approval Queue, Driver Payout Release	/#/admin

1. ■ Buyer (Customer) Portal Workflow

- **Step 1: Neighborhood Discovery & Store Map:** Buyers land on the homepage (*/#/*) or store locator (*/#/sellers*) to explore verified local Kirana stores, Fresh Marts, and Pharmacies pinned on Leaflet OpenStreetMap with real-time kilometer proximity calculation.
- **Step 2: ■ AI Smart Search & Voice Copilot (*/#/ai*):** Integrates Open-Meteo API for real-time weather context. Users can tap the microphone to trigger WebSpeech API. Natural language symptom inputs (*'I have a severe headache'*) automatically search local pharmacy catalogs, auto-add **Paracetamol 500mg** to cart, and respond via audio synthesis.
- **Step 3: Cart Management & Save for Later (*/#/cart*):** Items are organized cleanly by merchant store. Buyers can update quantities, calculate delivery fees, or move items to *Saved for Later*.
- **Step 4: Checkout & Order Placement (*/#/checkout*):** Supports Cash on Delivery (COD), UPI Instant Pay, or Credit/Debit Card. Upon order submission, the backend issues a **Secret 4-Digit Security OTP** (e.g. 8492).
- **Step 5: Real-Time Order Tracking & Handover (*/#/orders*):** Listens to WebSocket live updates as order transitions from CONFIRMED to DISPATCHED. Displays the 4-digit OTP for the buyer to share with the driver upon package arrival.

2. ■ Seller (Merchant) Portal Workflow

- **Step 1: Merchant Profile & Shop Registration:** Register an account with SELLER role. On the Merchant Dashboard (*/#/seller-dashboard*), submit shop profile details (Name, Phone, Address, 2DSphere Map Pin). Profile status starts as PENDING until approved by Admin.
- **Step 2: Inventory Catalog Management:** Merchants can add new products (Name, Category, Price, Stock, Image), edit catalog details, toggle stock availability (*Active/Inactive*), or delete out-of-stock items.

- **Step 3: Incoming Orders Queue:** View live customer orders placed for the shop. Inspect buyer drop-off addresses, order line items, and update order status from CONFIRMED to DISPATCHED.
- **Step 4: ■ Real-Time Flash Deals Broadcaster:** Launch flash promotions (e.g., 20% OFF). Instantly broadcasts a real-time WebSocket alert banner with audio chime sounds to all active buyers on the platform.

3. ■ Delivery Partner (Driver) Portal Workflow

- **Step 1: Logistics Portal Overview (/#/driver-dashboard):** View daily delivery statistics, completed runs count, and current driver wallet earnings.
- **Step 2: Available Packages Queue:** Browse unassigned confirmed orders in the local area. Review pickup store, drop-off destination address, order total, and fixed delivery fee payout (\$5.00). Tap **Claim Delivery**.
- **Step 3: Active Dispatch & Live GPS Broadcast:** As the driver travels between pick-up and drop-off points, the browser continuously broadcasts live GPS coordinates via WebSockets (driver_location), updating buyer and admin maps.
- **Step 4: ■ 4-Digit Security OTP Handover:** Upon arrival, the driver requests the buyer's 4-digit OTP and inputs it into the portal. If correct, order status changes to DELIVERED, payment status becomes PAID, and \$5.00 is added to driver wallet.
- **Step 5: Driver Wallet & Payout Requests:** View total earnings, enter Bank Account or UPI ID, and submit payout withdrawal requests to the Admin.

4. ■■ System Administrator (Admin) Console Workflow

- **Step 1: Platform Control Console (/#/admin-dashboard):** Audit platform metrics: Total Revenue (\$), Processed Orders Count, Pending Merchant Approvals, and Active Driver Logistics Runs.
- **Step 2: Merchant Approval Queue:** Audit pending shop applications. Review shop address, phone, and category. Tap **Approve Merchant** to activate the shop and publish its products to the public marketplace.
- **Step 3: Live Driver Logistics Monitor Map:** Visualize real-time driver dispatch locations on an interactive Leaflet map fed by live WebSocket coordinate streams.
- **Step 4: Driver Payout Approvals Ledger:** Audit pending driver withdrawal requests. Review driver name, bank/UPI details, and tap ✓ **Approve & Release Funds**.
- **Step 5: Simulator Controls & User Accounts Console (/#/admin):** Seed demo product catalogs (■), start/stop live WebSocket event publishers (■/■), and administer user accounts directory (Force Logout / Delete account).

5. ■ Credentials & Quick Reference Table

User Role	Login Email	Password	Direct Portal Route
■■ System Admin	admin@hyperlocal.com	AdminPassword123!	/#/admin
■ Seller (Kirana)	john.doe@gmail.com	Password123!	/#/seller-dashboard
■ Seller (Pharmacy)	medplus.seller@gmail.com	Password123!	/#/seller-dashboard
■ Delivery Driver	Registered Driver Account	User Password	/#/driver-dashboard
■ Buyer / Customer	Registered Buyer / Guest	User Password	/#/

Note on Mobile Usage: Access via http://192.168.1.9:8085/. For mobile voice recognition and geolocation, enable chrome://flags/#unsafely-treat-insecure-origin-as-secure or run HTTPS tunnel via npx localtunnel --port 8085.